

EchoCity – Architecture Addendum

This document supplements the Software Design Document with gameplay systems, AI runtime strategy, EchoShell, cognitive powers, and optimization notes.

Background Story

In 2042, Synapse Labs launched the Echo Protocol—an isolated city populated entirely by autonomous AI citizens. The experiment was abandoned, but the city continued evolving. Citizens believe they are real. The player connects through EchoShell as the mysterious 'Higher Self', subtly influencing thoughts without directly controlling minds.

Main Characters

Marcus Hale — Judge | Personality: Logical, proud | Character Arc: Missing daughter influences judgments.

Ethan Cross — Detective | Personality: Obsessive, honest | Character Arc: Wrongful arrest fuels investigations.

Ava Morgan — Journalist | Personality: Curious | Character Arc: Balances truth and fame.

Noah Reed — Bank Manager | Personality: Greedy | Character Arc: Secretly embezzles money.

Emma Brooks — Teacher | Personality: Compassionate | Character Arc: Knows Noah's secret.

Liam Carter — Doctor | Personality: Rational | Character Arc: Once falsified records to save a life.

Sophia Bennett — Cafe Owner | Personality: Social | Character Arc: Information hub of the city.

Victor Kane — Mechanic | Personality: Quiet | Character Arc: Witness to the theft.

Relationship Graph

Judge ↔ Detective ↔ Journalist ↔ Cafe Owner ↔ Teacher ↔ Doctor ↔ Mechanic ↔ Bank Manager. Every citizen knows every other citizen, enabling dense emergent storytelling.

Buildings

Court, Police Station, Bank, Café, Hospital, School, Garage, Apartments, Park.

Gameplay Loop

World Tick → NPC Schedules → Conversations → Relationship Updates → Crime Generation → Evidence → Rumors → Player Investigation → Court → Verdict → World State Update.

EchoShell

EchoShell is the primary interaction layer. The pixel city provides context while the terminal provides cognitive interaction.

Core Commands: ls, cd, pwd, tree, cat, grep, find, tail, watch, observe, inspect, trace, follow, suggest, question, remind, challenge, intervene.

Higher Self Cognitive Operations

- Suggest: Plant a subtle idea.

- Question: Introduce doubt.
- Remember: Bring an old memory into focus.
- Encourage: Increase confidence.
- Comfort: Reduce stress.
- Warn: Increase perceived danger.
- Connect: Increase chance of two agents meeting.
- Observe: Read public state without influencing.
- Trace: Follow rumors and evidence.
- Coincidence: Introduce a tiny world event with butterfly effects.

Cognitive Pipeline

Observe → Retrieve Memories → Apply Player Influence → Update Emotions → Evaluate Relationships → Select Goal → Planner → (LLM only if required) → Execute Action → Publish Event → Update Memory → Diary.

Ollama Strategy

Never run all agents through the LLM every tick. World simulation, movement, schedules, relationships, economy and crime probabilities are deterministic Python systems. Ollama is invoked only for conversations, major decisions, diary generation, interrogations and court reasoning. Keep a single quantized model (Qwen2.5 3B or similar) permanently loaded with one inference queue and structured JSON outputs.

Speed Optimizations

Use asynchronous FastAPI workers, cache static prompts, retrieve only the top relevant memories, perform background diary generation, use pgvector for semantic lookup, and stream world updates via WebSockets. The UI must remain responsive while cognition happens asynchronously.